

Recovering the Unpublished Projection Parameters of Bulgaria's 1970 Coordinate System

A reconstruction of zones K3, K5, K7 and K9 from 276,003 control points over a 20 km-buffered extent — what is recoverable, what provably is not, and where the official transformation itself breaks.

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276,003 matched points · 4 zones · numpy + PROJ 9.7

ABSTRACT

Since 2010, Bulgarian cadastral and topographic work has been carried out in the Bulgaria Geodetic System 2005 / Cadastral Coordinate System 2005 (BGS2005/CCS2005, EPSG:7801) — a Lambert Conformal Conic projection on GRS80 with fully published parameters. A large body of legacy survey and cadastral data, however, remains referenced to the earlier “Coordinate System 1970” (KS1970), realized in four Lambert zones — K3, K5, K7 and K9 — on the Pulkovo 1942(58) datum / Krasovsky 1940 ellipsoid. Neither the zone parameters nor the BGS2005 → KS1970 transformation constants were ever officially published, and the cadastre agency's own tool is closed source. Here we recover both a portable analytic approximation and a survey-grade interpolation grid for all four zones — from 276,003 real coordinate pairs generated by the agency's own software over an extent buffered 20 km beyond the zone boundaries — using nothing but numpy and PROJ. Alongside the achieved accuracy we report two findings that change how such reconstructions should be presented: the conic parameters are not recoverable from planar correspondences (only the cone constant is), and the agency's transformation is itself discontinuous exactly along the 24°E meridian in the two western zones.

01 Introduction

Two coordinate systems are in daily parallel use in Bulgarian survey and GIS practice. BGS2005/CCS2005 has been the national system since 2010, with published and fully determined parameters (EPSG:7801). Alongside it, a large volume of older cadastral plans, control network points and utility infrastructure records remain tied to the earlier Coordinate System 1970. Practitioners routinely

need to move coordinates between the two — checking an old boundary marker against a modern cadastral map, digitizing archive plans, or bringing a long-established utility network into a current GIS.

The Agency for Geodesy, Cartography and Cadastre (AGKK) provides a free but closed, Windows-bound tool — BGSTrans — for such conversions. No scriptable or open-source implementation with published parameters exists, which makes automated and batch workflows effectively impossible without reverse-engineering the parameters.

02 Two Bulgarian coordinate systems

BGS2005 / CCS2005 (EPSG:7801)

GRS80 ellipsoid; Lambert Conformal Conic with two standard parallels; central meridian 25°30'E; standard parallels 42°00' and 43°20'N; latitude of origin 42°40'04.35"N; false easting 500,000 m; false northing 4,725,824.3591 m. Adopted in 2010 as a single national zone.

COORDINATE SYSTEM 1970 (KS1970)

The predecessor system, built on the Pulkovo 1942(58) datum on the Krasovsky 1940 ellipsoid, divided into four Lambert Conformal Conic zones by compass quadrant — **K3** (northwest), **K5** (southeast), **K7** (northeast) and **K9** (southwest). The `.prj` files AGKK's own tool writes name the datum and the projection *type*, and then carry no numeric parameter at all.

03 The information gap

Published sources confirm the absence of official figures: a Geomedia.bg analysis comparing KS70 with the Russian KS63 states explicitly that KS70's parameters “have not been published to date” — decades after the system's introduction. Independent confirmation comes from an existing open-source project ([bojko108/transformations](https://github.com/bojko108/transformations), GitHub), which supplies a precomputed interpolation grid derived by sampling AGKK's official software rather than a formula, because its author had no published formula either. That motivates the approach here: recovery directly from data, as the only currently workable open-source path.

04 Data

For each of the four zones we have points in BGS2005/CCS2005 and the same points transformed into that zone of KS1970 by AGKK's official software. Each pair is joined by a shared integer id, giving exact correspondences rather than

manually surveyed control points. The tool's own report declares a plane accuracy of 0.14 m and a zone width of 3 degrees.

Sampling covers each zone's extent **buffered 20 km outward**. That matters in practice: a grid sampled exactly to the zone boundary stops working precisely where it is most often needed — along the boundaries between zones, and for features that cross them. The buffer carries coverage past the boundary and makes neighbouring zones overlap in data coverage, without changing the zones themselves. Measured on K9's own official boundary vertices, the un-buffered grid covers 9 of 126 of them (7%); the buffered grid covers all 126.

Every zone is sampled **twice** — on a 1 km and on a 2 km lattice. The two lattices are offset from each other by exactly 500 m, so every 2 km point falls at the *centre* of a 1 km cell and none coincides with a 1 km node (verified: zero coordinate collisions). Every model is therefore built on the 1 km set and scored on the 2 km set. For the interpolating model that distinction matters a great deal: a random hold-out of a single lattice leaves test points sitting on the *edges* of triangles built from their own immediate neighbours, which flatters the result. Cell centres are the worst possible position for piecewise-linear interpolation, so the reported errors are an upper bound rather than a favourable sample.

Table 1 — control points and zone extents				
ZONE	QUADRANT	1 KM POINTS	2 KM POINTS	AREA (KM²)
K3	northwest	47,804	11,939	23,695
K5	southeast	56,594	14,159	31,891
K7	northeast	53,912	13,478	30,378
K9	southwest	62,497	15,620	37,333

The zone extents come from a separate layer of their official boundaries. The four polygons **tile the country rather than overlapping**: their areas sum to 123,297 km² against a union of 123,281 km². The 20 km-buffered sampled extent runs from 206,222 to 780,705 m easting and 4,539,859 to 4,928,413 m northing in CCS2005.

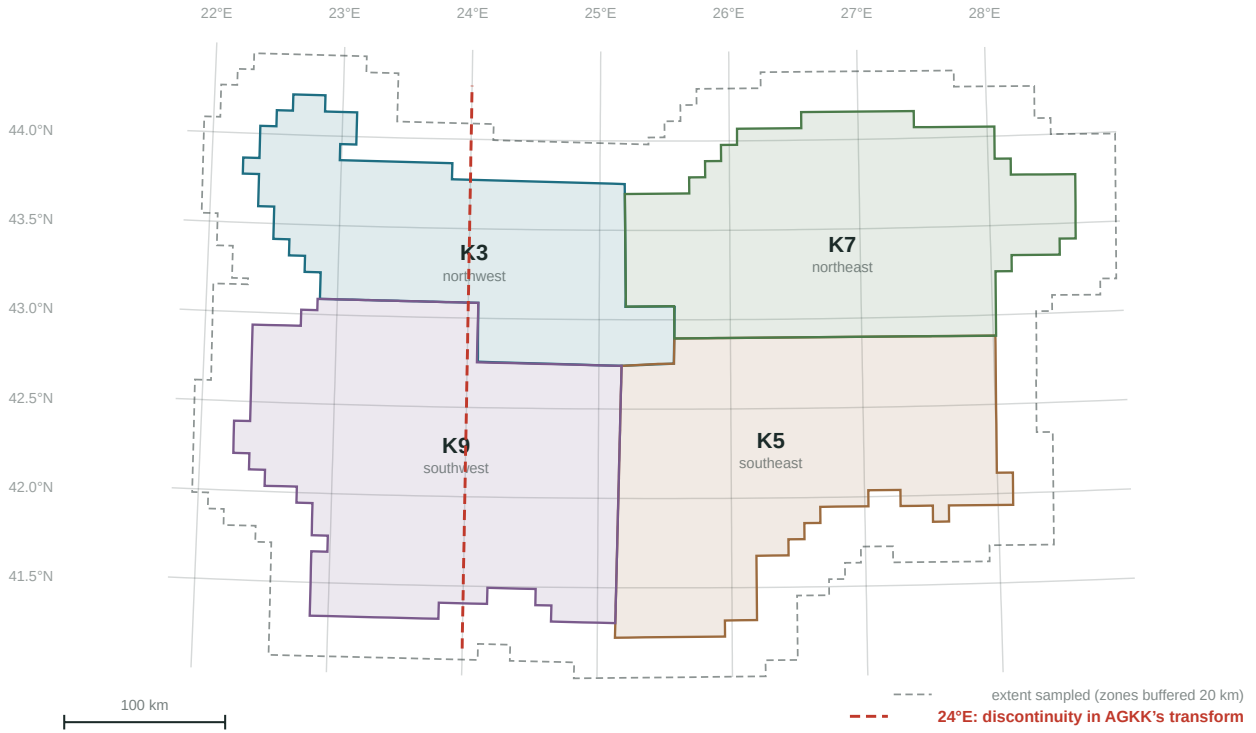


Figure 1. The four KS1970 zones, drawn from their official boundaries, with the dashed outline showing the extent the control points are actually sampled over — the union of the zones buffered by 20 km. Every zone is labelled, so colour only duplicates the label. The red dashed line is the 24°E meridian — the line along which AGKK's transformation is not smooth in K3 and K9 (section 08).

05 Method

5.1 CLOSED-FORM (ANALYTIC) MODEL

Because the datum transformation to the target ellipsoid is also unknown, we avoid fitting a full 7-parameter geocentric Helmert transformation — ill-conditioned at the scale of a single zone and largely redundant with a final planar correction. Instead, geographic coordinates on GRS80 (obtained by inverting the fully known EPSG:7801 definition) are passed through an ellipsoidal Lambert Conformal Conic projection (Snyder, 1987), followed by a planar similarity (rotation θ , scale s , translation t_x , t_y) fitted to the observed KS1970 coordinates.

Which parameters are fitted matters. As section 07 shows, the usual triple (λ_0 , ϕ_1 , ϕ_2) is not determinable from planar correspondences. We therefore fit the identifiable 5-parameter form directly — the cone constant n together with θ , s , t_x , t_y , with λ_0 pinned to 25.5° by convention — using nonlinear least squares (Levenberg–Marquardt).

5.2 GRID MODEL (TRIANGULATED INTERPOLATION)

The control points form a regular lattice clipped to the zone's sampled extent, so the triangulation is done cell-wise rather than by Delaunay. This is not merely faster but more faithful: Delaunay triangulates the convex hull, and the extents are not convex, so Delaunay would span the concavities with long, thin triangles interpolating across territory that was never sampled. Cell-wise coverage spans the sampled region and nothing more. PROJ's `tinshift` operator (core since 7.2) interpolates barycentrically between the vertices — in both directions, from a single JSON file. Accuracy is measured through PROJ's own engine rather than a reimplementation.

06 Results

RMSE below is the root-mean-square planar deviation between computed and observed point. All figures are out of sample, on points that take no part in the fitting.

Table 2 — analytic model: identifiable parameters and out-of-sample accuracy							
ZONE	CONE CONSTANT N	TANGENT PARALLEL	ROTATION (")	SCALE (PPM)	RMSE (M)	P95 (M)	MAX (M)
K3	0.687723149	43.450145°	+5704.9	−6.55	0.172	0.329	0.445
K5	0.675170794	42.467415°	−2161.3	−3.05	0.213	0.401	0.641
K7	0.689029005	43.553295°	−1812.4	+1.41	0.190	0.352	0.494
K9	0.672869214	42.288900°	+5041.6	−8.81	0.139	0.282	0.497

These figures are of the order of AGKK's own declared 0.14 m plane accuracy, ranging from 0.14 to 0.21 m by zone. Fit-set and test-set RMSE agree to three decimals in every zone, so the residual is model bias, not overfitting: a single global conic cannot express the local structure of the historical 1970 network adjustment. Against the earlier, un-buffered sampling the same cone constants reproduce to five decimals while the RMSE rises — as expected, since one global model now has to cover 20 km more territory in every direction.

Table 3 — grid model: out-of-sample accuracy								
ZONE	VERTICES	TRIANGLES	SIZE	COVERED	MEDIAN (MM)	P95 (MM)	MAX (MM)	> 1 MM
K3	47,804	94,569	4.54 MB	11,810 / 11,939	0.535	0.544	79.3	36
K5	56,594	112,120	5.39 MB	14,025 / 14,159	0.137	0.139	0.139	0
K7	53,912	106,798	5.13 MB	13,343 / 13,478	0.607	0.615	0.616	0
K9	62,497	123,915	5.96 MB	15,497 / 15,620	0.262	0.266	182.3	61

Uncovered points fall just outside the built lattice's outline; they are excluded rather than extrapolated. Round-trip error, forward then inverse, is under 3 nm. All 97 residuals above 1 mm in the entire project lie on a single line, which is the subject of section 08.

07 What the data does not determine

It is tempting to fit the textbook triple $(\lambda_0, \varphi_1, \varphi_2)$ alongside the similarity and publish the fitted $\lambda_0/\varphi_1/\varphi_2$ as “the recovered zone parameters”. That would be wrong. Those seven parameters are not identifiable from planar correspondences:

- changing λ_0 rotates the projection plane about the cone apex, which the similarity's own rotation θ absorbs exactly — only $\theta - n \cdot \lambda_0$ is determined;
- changing (φ_1, φ_2) at constant cone constant n changes only the projection's overall scale factor, which the similarity's scale s absorbs exactly.

Table 4 — zone K3: three fits from widely separated starting points

START Λ_0	FITTED Λ_0	Φ_1	Φ_2	SCALE-1 (PPM)	CONE N	RMSE (M)
23.0°	23.119220°	43.347°	43.554°	-4.9	0.687723149	0.1720
25.5°	24.353738°	50.062°	36.583°	+6936.3	0.687723149	0.1720
27.5°	25.341094°	47.275°	39.541°	+2268.2	0.687723149	0.1720

FINDING 1

Parameters differing by **whole degrees** and by nearly 7000 ppm of scale produce coordinates agreeing to **under one micrometre**, and agree on n to nine significant figures. The conic shape recoverable from this kind of data is exactly one number — the cone constant — plus a planar similarity.

Table 2 therefore reports n , not λ_0 and the standard parallels. The tangent parallel there is a reparameterization of n ($\arcsin n$), and λ_0 is a convention. Neither should be quoted as an AGKK parameter.

08 A discontinuity in the official transformation

Along any straight row of the lattice, a smooth projection produces a second difference of a couple of millimetres per km^2 — that is just the conic's own curvature. In K3 and K9, however, a localized spike two orders of magnitude higher appears.

How this is aggregated is critical. The kink sits on a meridian, and a meridian is not a column of this projection's grid: over a zone's ~ 250 km of latitude the easting of the 24°E meridian drifts by about 4 km. Bin by column and the kink is smeared across eight columns, present in under half the rows of each, and a median hides it completely. Bin by longitude and every row's kink lands in the same bin.

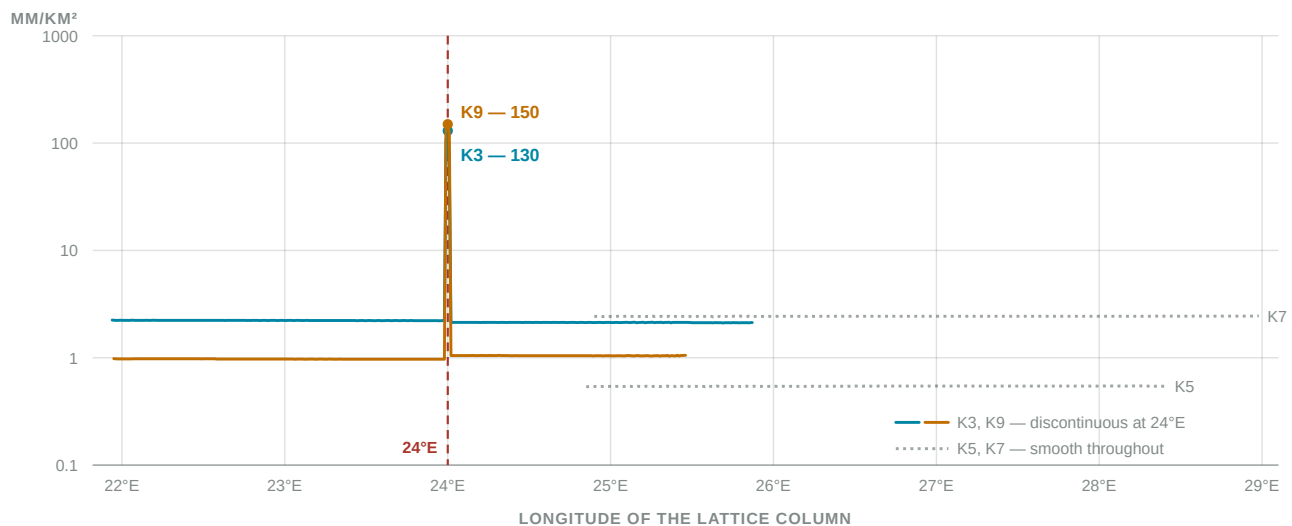


Figure 2. Second difference of AGKK's output along the lattice rows, aggregated into longitude bins, in mm/km²; logarithmic scale. K5 and K7 are flat across their full width (0.54 and 2.43 mm/km²). K3 and K9 are equally flat except for a single spike reaching 130 and 150 mm/km² — exactly at 24.0000°E.

Table 5 — the discontinuity, by zone

ZONE	SEAM LONGITUDE	PEAK	TYPICAL CURVATURE	RESIDUALS > 1 MM
K3	24.0000°	130 mm/km ²	2.2 mm/km ²	36
K9	24.0000°	150 mm/km ²	1.0 mm/km ²	61

FINDING 2

This is a property of AGKK's transformation, not of our data or of the triangulation: building the grid from one set and scoring it on the other reproduces the discontinuity at the same longitude, and the earlier, independent sampling shows it in the same place. It is almost certainly a panel boundary in the tool's own internal correction grid. K5 and K7 show nothing comparable.

The consequence: **all 97 residuals above 1 mm in the entire project lie on this line.** The grid reproduces the kink exactly at the sampled points but must interpolate straight across it in the one cell that straddles it. Excluding those points, each zone's out-of-sample RMSE equals its median: 0.535, 0.137, 0.607 and 0.262 mm. No closed-form model can represent the kink at all.

09 Practical use

We provide a reproducible open-source workflow: PROJ pipeline strings, `tinshift` JSON grids (one per zone, both directions via `+inv`), a QGIS-importable Custom CRS (WKT) wrapping the analytic model, and command-line and Python recipes. The only dependencies are `numpy` and `pyproj`; the nonlinear least squares and the triangulation are implemented directly.

EXAMPLE — PROJ PIPELINE (GRID, ZONE K3)

```
echo "300000 4800000" | gdaltransform -ct \  
"+proj=pipeline +step +proj=tinshift +file=bg_k3_tinshift.json"
```

Recommendation: the grid model for cadastral-accuracy work; the analytic formula for portability, for extrapolation beyond the sampled territory, or for embedding where a grid file cannot be distributed.

10 Limitations

This reconstruction is not an official or legally valid substitute for AGKK's BGSTrans — it is an empirical recovery from a sample of correspondences, valid within each zone's sampled territory. The conic parameters in the results are a convention, not a recovered definition (section 07). The target datum (Krasovsky 1940 / Pulkovo 1942(58)) is taken from the files' metadata rather than independently derived — though neither method needs an explicit datum-shift model, since both work from real correspondences. Finally, the 24°E discontinuity is inherited from AGKK's tool: survey work crossing that meridian in K3 or K9 should expect roughly 0.1 m of disagreement there whichever method is used.

11 Conclusion

We present two complementary, freely reproducible artifacts that recover KS1970's K3, K5, K7 and K9 zones at practical (analytic model, 0.14–0.21 m) and at geodetic (grid model, 0.14–0.61 mm) accuracy, using open-source tooling alone. Equally important is what we do *not* claim: what such data recovers is the cone constant, not the standard parallels, and the official transformation is itself not smooth everywhere. This fills a genuine gap in Bulgarian GIS and geodetic practice, pending any future official publication of the true parameters.

REFERENCES

1. Geomedia.bg — “The Bulgarian Coordinate System KS70 and KS63” (comparative analysis).
2. Geomedia.bg — “Accuracy of Cadastral Parcel Areas, KS1970 vs. BGS2005.”
3. bojko108/transformations, GitHub — interpolation grids for BGS2005 ↔ KS1970 K3/K5/K7/K9.
4. EPSG Geodetic Parameter Registry — EPSG:7801 (BGS2005/CCS2005), EPSG:7798, ESRI:4179.
5. PROJ / OSGeo — `tinshift` operator documentation, PROJ $\geq$ 7.2.
6. Snyder, J. P. (1987) — *Map Projections: A Working Manual*, USGS Professional Paper 1395.
7. AGKK — BGSTrans, free coordinate-transformation tool (cadastre.bg).

This paper and its accompanying data, scripts and parameters are an independent, reproducible reconstruction built with open-source tooling (numpy, PROJ/pyproj, GDAL). It is not an official AGKK product. Free to republish with attribution. © 2026 Todor Lubenov.